LO #2

1. Show that Coulomb's Law is a solution to the integral form of Gauss' Law. I.e., show that the following are equivalent for a charge of strength q placed at the origin.

$$\vec{E}_{1,2} = \frac{\vec{F}_{1,2}}{q_1} = \vec{R}\frac{q_2}{4\pi\varepsilon|\vec{R}_{1,2}|^3}$$

$$\oint_S \vec{D}\cdot d\vec{S} = Q_{\text{enclosed}}$$

2. An infinitely long cylindrical shell of charge is centered on the z-axis and extends between $\rho = 1\text{m}$ and $\rho = 3\text{m}$. The volume charge density within this shell is uniform and is equal to ρ_v (C/m^3). Use the integral form of Gauss' Law to find the electric field intensity everywhere. Note: there are three regions to consider for ρ.

3. A distribution of electric charge in the shape of a cube of dimensions 1cm × 1cm × 1 cm gives rise to the electric field intensity $\hat{\boldsymbol{r}}\frac{3}{r^2}$ V/m. The charge of the cube is centered at the origin. What is the total enclosed charge in the cube?

4. An infinite line of charge of uniform charge distribution -2.1mC/m exists along the z-axis (i.e., $x = y = 0$). The surrounding media is plastic, which is well-characterized as a homogeneous medium having relative permittivity (ε) equal to 2.5. What is the electric flux density everywhere? Please express your answer as a numerical value as much as possible.